

What is software ?

Software: It usually corresponds to routines, programs, symbolic language, data which control the functioning and directs the operation of hardware.

□ It cannot be touched, has no weight, no material or energy but when it is executed properly it performs a specific task.

In other words, software can be considered to be an *abstract-ware* which when executed properly performs a specific task.

Many times it is defined as group of programs which perform a specific task.

Program is usually defined as the set of instructions to perform a specific task



What is Software?

Software is:

- (1) instructions (computer programs) that when executed provide desired features, function, and performance;
- (2) data structures that enable the programs to adequately manipulate information, and
- (3) descriptive information in both hard copy and virtual forms that describes the operation and use of the programs.

Source: Roger S. Pressman, Bruce R. Maxim, "Software Engineering: A Practitioner's Approach", 8th edition, McGraw-Hill, 2019.

Some software disasters

Therac - 25

The **Therac-25** was a computer-controlled radiation therapy machine produced by Atomic Energy of Canada Limited(AECL) in 1982 after the Therac-6 and Therac-20 units (the earlier units had been produced in partnership with CGR of France).



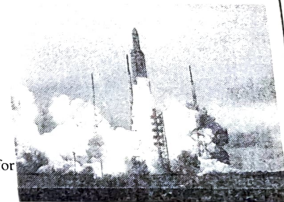
- 6 known accidents resulting in death or serious injury (during June 1985 – January 1987) in which patients were given massive overdoses of radiation.

It was because of concurrent programming errors, the machine sometimes gave its patients radiation doses that were hundreds of times greater than normal, resulting in death or serious injury.

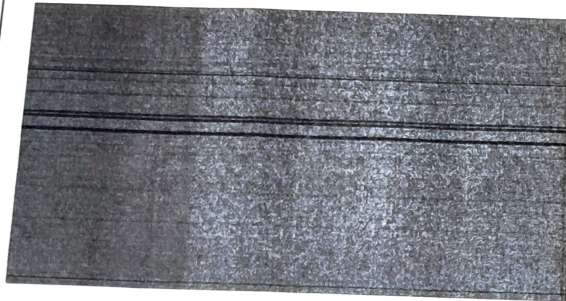
- Software was adapted from earlier models

Ariane 5

- In 4 June 1996, Ariane 5 launcher exploded about 40 seconds after launch on maiden flight
- Rocket and cargo worth about \$500 million
- Due to failure of Inertial Reference Systems
- Failure caused by software trying to convert a 64-bit floating point value for a horizontal velocity into a 16-bit integer
- Much of the software used in Ariane 5 was same as that used in Ariane 4



But, the horizontal velocity on Ariane 4 never reached as high a value as on Ariane 5, so it was not a problem on Ariane 4



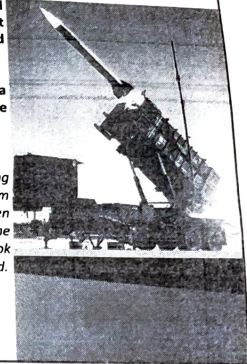
Source: www.youtube.com

The Patriot Missile

During the Gulf War, 28 U.S. soldiers were killed and almost 100 others were wounded when a nearby Patriot missile defense system failed to properly track a Scud missile launched from Iraq.

The cause of the failure was later found to be a programming error in the computer embedded in the Patriot's weapons control system.

The software problem led to an inaccurate tracking calculation that became worse the longer the system operated. At the time of the incident, the [Patriot] had been operating continuously for over 100 hours by which time the inaccuracy was serious enough to cause the system to look in the wrong place [in the radar data] for the incoming Scud.



Taurus

Taurus (Transfer and Automated Registration of Uncertified Stock) was a program that set out to transfer settlements of London Stock Exchange shares from transmission of paper share certificates to an automated system. TAURUS was intended to reduce the time taken and the cost of settlement, increasing convenience and reducing settlement risk.

Project cancelled in 1993 after more than 5 years of development.

About £75m was the project cost, about £400m was the loss to customers.

Causes: Tried to do too much in one go; tried to simplify(!), computerise sale of stocks and shares and make process paperless

- Chose expensive new system instead of modifying existing one
- Used committees with conflicting interests to specify requirements.
- Had two competing consultancies (Andersens and Coopers & Lybrands) working on the project

Denver Baggage Handling System

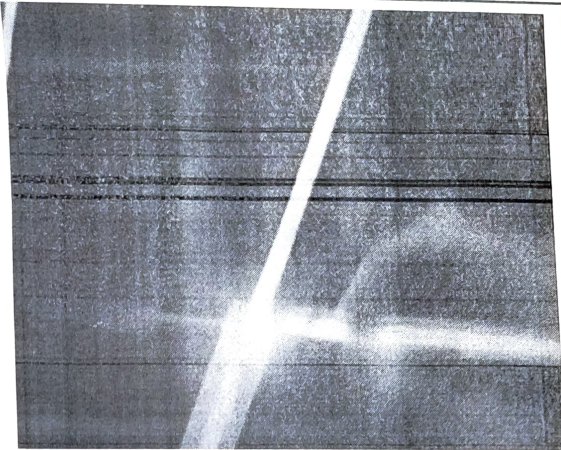
• Complex system involving 300 computers

• Project overran badly

delayed opening of airport by 16 months (opened around Feb 1995)

The delay added approximately \$560M (USD) to the cost of the airport and became a feature article in Scientific American titled "Software's Chronic Crisis".

The product was extremely buggy



Source: History Channel

Some terminologies

- **Science:** The observation, identification, description, experimental investigation and theoretical explanation of phenomena. It makes use of scientific method which involves careful observation of phenomena, the formulation of hypothesis, conducting one or more experiments to test the hypothesis and the drawing of a conclusion that confirms or modifies the hypothesis.
- **Technology:** It is the making, modification, usage and knowledge of tools, machines, techniques, crafts, systems and methods of organization, in order to solve a problem, improve a pre-existing solution to a problem, achieve a goal, handle an applied input/output relation or perform a specific function. It can also refer to the collection of such tools, including machinery, modifications, arrangement and procedures.
- **Engineering:** It is the application of scientific, social and practical knowledge in order to design build and maintain structure, machines, devices, systems, materials and processes. It may encompass using insights to conceive, model and scale an appropriate solution to a problem or objective.

Software errors, software faults and software failures

- **Failure:** the inability of a system or component to perform its required functions within specified performance requirements.
- **Fault/Defect/Bug:** a generic term that means something imperfect (*incorrect step, process or data definition*) or does not work as expected
- **Error:** A mistake or flaw that may produce an incorrect or unexpected result.

When people make mistakes while coding, people usually also call these mistakes bugs.

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Terminologies

- ❖ An error of omission (can result in fault of omission) occurs when there is a mistake of leaving/missing something which should have been present in the representation.
- ❖ A fault of commission occurs when we enter something into a representation that is incorrect.
- ❖ Faults of omission are more difficult to detect and resolve.

Incident—When a failure occurs, it may or may not be readily apparent to the user (or customer or tester). An incident is the symptom associated with a failure that alerts the user to the occurrence of a failure.

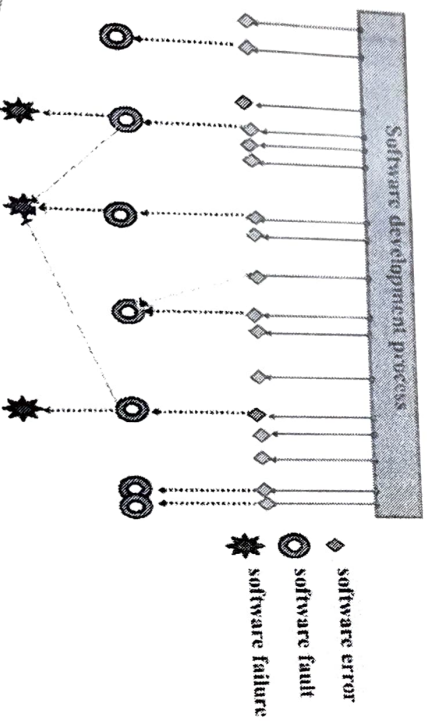
Test—A test is the act of exercising software with test cases to find failures or to demonstrate correct execution.

Terminologies

Test case - A complete test case usually contains a test case identifier, a brief statement of purpose (e.g., a business rule), a description of preconditions, the actual test case inputs, the expected outputs, test steps (a description on how to execute tests), a description of expected postconditions, and an execution history.

- ❖ The execution history is primarily for test management use—it may contain the date when the test was run, the person who ran it, the version on which it was run, and the pass/fail result.

Software errors, software faults and software failures



Software errors, software faults and software failures

- Not all software errors become **software faults**, in some cases, the software error can cause improper functioning of the software.
- In many other cases, erroneous code lines will not affect the functionality of the software as a whole.
- A failure is said to occur whenever the external behaviour of a system does not conform to that prescribed in the system specification.

Errors can cause fault but not all errors can lead to fault.

Similarly, when the faulty code is executed it may result in the failure of the entire system but all faults may not lead to failure.

DEMONSTRATION-ORIENTED TESTING

- ❖ In this phase, there was an introduction of software testers independent of development activity.
- ❖ Their main aim was to show to customer/users that the software really works.
- ❖ Although, this phase was much advanced than the initial phase of debugging, yet the approach was still positive testing only.
- ❖ The approach was oriented towards demonstration that the software could do something which was expected by the customer (s) ~~(s)~~ ^{(s)/user(s)}.
- ❖ Test cases were generally derived from the requirement statements which were oriented towards successful demonstration.

DESTRUCTION-ORIENTED TESTING

- ❖ This approach was the basis of Glenford Myer's theory of software testing.
- ❖ As per this approach, it was not sufficient to only test the software positively but users must also be protected from any conceivable failure of application.
- ❖ The tester's responsibility changed from 'proving that software works under normal conditions' to 'proving that software does not fail at some abnormal instances'.
- ❖ Often the testers were too imaginative in breaking the software, and defects for which there were no feasibility of happening were reported as defects.
- ❖ This phase was quite frustrating to software developers - testers were considered as demons and testing was considered as a hurdle to be passed before delivering the application to the customer.

EVALUATION-ORIENTED TESTING

- ❖ Evaluation-oriented testing is executed nowadays at many places of software development where the product as well as process of software development is evaluated.
- ❖ It corresponds to the testing process definition where software is evaluated against some fixed parameters derived from quality factors (test factors).
- ❖ Quality factors/test factors are introduced in this phase as a part of customer requirements.
- ❖ It is believed that there is no possibility of software without any defect, but some level of defect may be acceptable to the customer.
- ❖ This approach also refers to level of confidence given to customer that application will work as expected by the user.
- ❖ The confidence level is determined and application is evaluated against it.
- ❖ Confidence-level expectations are linked with cost of testing.

PREVENTION-ORIENTED TESTING

- ❖ Prevention-based testing is done in some highly mature organizations while for many others, the concept is still utopia.
- ❖ Testing is considered as a prevention activity where process problems are used to improve it so that defect-free products can be produced.
- ❖ Every defect found in testing is considered as process lacunae, and efforts are initiated to improve the process of development.
- ❖ This reduces dependency on testing as a way to improve the quality of software.
- ❖ This helps in reducing the cost by producing right product at the first time.